Yiping Li

Email: yipingli@ucmerced.edu; yipingl1024@gmail.com | Phone: 206-302-8770 | GitHub | Google Scholar | yipingli.dev

RESEARCH INTERESTS

My research spans LLM post-training and multi-agent systems, efficient inference, and AI for science. My current work includes reinforcement learning and on-policy distillation, AI hardware simulation and multi-agent KV-cache compression, and quantum circuit synthesis and reasoning.

EDUCATION

University of California, Merced, Merced, CA — 08/2025 – Expected 05/2029
- Ph.D. in **Electrical Engineering and Computer Science** (Advisor: Dr. Wan Du)

University of Illinois Urbana-Champaign, Champaign, IL — 08/2022 – 12/2023
- Master of **Computer Science**

University of Washington, Seattle, WA — 10/2017 – 06/2022
- B.S. in **Applied & Computational Mathematical Sciences (ACMS)** & **Psychology**

SKILLS

Methods: on-policy distillation, knowledge distillation, RL post-training, preference optimization (DPO), KV-cache compression, LLM evaluation, LoRA/PEFT, model merging, federated learning, RAG

Languages and tools: Python, Java, SQL; PyTorch (DDP), Hugging Face Transformers and Accelerate, TensorFlow, verl, Unsloth, NumPy, Pandas, scikit-learn; AWS, Linux, Git

PUBLICATIONS AND PREPRINTS

- **Li Y**, Zhao F, Hou Z, An Z, Du W. Demystifying On-Policy Distillation: Where Does the Teacher's Advantage Live? Preprint, 2026. arXiv: 2610.19145.
- **Li Y**, An Z, Du W. Compressing KV-Cache Communication in Multi-Agent Systems with Collective Compensation. Under review, ICLR 2027. arXiv: 2604.13349.
- **Li Y**, An Z, Du W. Sanzaru: Benchmarking Protocol Formation under Epistemic & Communication Asymmetry. Under review, ICLR 2027. Co-first author. arXiv: 2610.15163.
- Nakshbandi D, An Z, **Li Y**, Giri S, Du W. Neither True nor False: Do LLMs Represent Truth-Value Gaps in Plural Predication? Preprint, 2026. arXiv: 2610.14438.
- Li J, Zhang Y, **Li Y**, Gong X, Wang W. FedSparse: A Communication-Efficient Federated Learning Framework Based on Sparse Updates. Electronics, 13(24):5042, 2024.
- Li J, Zhang Y, **Li Y**, Gong X, Wang W. HybridCom: Improve Federated Learning Efficiency on Unstable Data. IEEE ICC, pp. 2622-2627, 2024.
- Li J, Zhang Y, **Li Y**, Gong X, Wang W. Gradient Calibration for Non-IID Federated Learning. ACM Workshop on Data Privacy and Federated Learning Technologies for Mobile Edge Network, pp. 119-124, 2023.
- Li J, Zhang Y, **Li Y**, Gong X, Wang W. DCD: A New Framework for Distillation Learning with Dynamic Curriculum. IEEE HPCC/DSS/SmartCity/DependSys, pp. 928-935, 2022.

RESEARCH EXPERIENCE

University of California, Merced — 08/2025 – Present

Research Assistant, Advised by Dr. Wan Du

Collective compensation for multi-agent KV communication
- Developed Orthogonal BackFill (OBF) to collectively recover complementary information from discarded KV values and fold it into the retained cache without increasing its size.
- Reduced per-agent KV-cache relay volume by approximately 4.7x versus full-cache relay. At matched relay budgets, OBF improved accuracy over eviction alone in 31 of 42 comparisons across seven benchmarks and Qwen3 models from 4B to 14B.

Benchmarking multi-agent coordination under communication asymmetry
- Developed SANZARU, a benchmark of 120 bomb-defusal puzzles across seven families, using asymmetric knowledge and communication channels to evaluate multi-agent LLM coordination.
- Found a large coordination gap: GPT-5.6-Luna solved 90% of puzzles with full information but 41% as a three-agent team. Across all evaluated teams, communication failures accounted for about 60% of 771 losses.

Demystifying On-Policy Distillation

- Developed token-flip experiments to test where a teacher improves student reasoning, replacing student tokens with teacher choices and comparing outcomes against student resampling without updating model weights.
- Found that the teacher's advantage is spread across the reasoning trajectory, with small gains accumulating through repeated token flips. Gains vary across positions, making some flips more useful than others.

LLM Infrastructure and AI Hardware Simulation

- Developing a page-level LLM inference simulator to evaluate hypothetical chip designs, including HBF-based memory architectures.
- Modeled GPU kernels, L2 caching, and weight/KV traffic across memory tiers to diagnose bottlenecks and compare placement and prefetching policies.

LLMs for Quantum Computing

- Investigating encoder-based approaches to fault-tolerant quantum circuit synthesis.
- Auditing and correcting Qiskit QuantumKatas, then adding paired multiple-choice tasks to distinguish code-generation failures from failures in underlying quantum reasoning.

Joint Laboratory of BUPT and ChuangCache 08/2021 – 03/2025

Research Assistant (Remote), Advised by Dr. Yuchao Zhang

Designed core algorithms, implemented experiments, and built a shared federated-learning evaluation framework supporting the four publications listed above.

Communication Efficiency in Federated Learning

- Designed a Resource Optimization Proximal (ROP) loss enforcing sparse client updates, cutting communication up to 24% at near-unchanged accuracy.
- Used historical gradients to weight sparsity penalties by parameter importance, preserving important updates while reducing communication of less important ones.

Gradient Calibration in Federated Learning

- Built Federated Gradient Tailor (FGT), improving non-IID MNIST/CIFAR-10 accuracy by 3.2-5.8 percentage points over FL without gradient calibration and reaching baseline accuracy in 2-6x fewer communication rounds.
- Authored the formal convergence proof for FGT.

Federated Learning Efficiency on Dynamic Non-IID Data

- Developed HybridCom to prioritize clients whose changing data distributions were not yet reflected in the global model, combining local performance and server-side gradient signals to improve accuracy by approximately 1.3% over evaluated baselines.

Integration of Knowledge Distillation and Curriculum Learning

- Developed DCD to adjust the order of training samples during knowledge distillation using feedback from the teacher and student.
- Improved accuracy over knowledge distillation with random sampling by 1.7%, 2.5%, and 1% on CIFAR-10, CIFAR-100, and CINIC-10, respectively.

PROFESSIONAL EXPERIENCE

Brix AI, Irvine, CA 07/2024 – 05/2025

Machine Learning Engineer

- Designed a semantic labeling taxonomy to turn recruiting criteria, such as big-tech-company experience and university tiers, into searchable resume attributes.
- Built an LLM pipeline using RAG to align extracted keywords with existing labels and tool calls to save new keywords when no suitable match existed.
- Built backend workflows for resume analysis and job-description generation using OpenAI APIs, function calling, and file search, reducing token usage by 20%.

Orbex Labs, Remote 03/2024 – 07/2024

Machine Learning Intern

- Compared YOLOv8 configurations for runtime and cost, then developed a word-frequency document classifier that outperformed the YOLOv8 baseline at lower cost.
- Replaced an oriented bounding-box model with RANSAC-based rotation and scale correction, increasing text-extraction accuracy from 33% to 99%.

Baidu, Inc., Beijing, China 08/2020 – 03/2021

Machine Learning Intern

- Built a deepfake-detection evaluation pipeline with dataset balancing and automated ROC/F1 reporting, contributing to a 2% accuracy gain.
- Developed rule-based filtering for adversarial text, blocking over 600,000 harmful comments that bypassed BERT.